

Linear Algebra – MAT 2610

Section 5.2 (The Characteristic Equation)

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Theorem 3

Let A and B be $n \times n$ matrices

- a. The matrix A is invertible if and only if $\det(A) \neq 0$
- b. $\det(AB) = \det(A) * \det(B)$
- c. $\det(A^T) = \det(A)$
- d. If A is triangular, then $\det(A)$ is the product of the entries on the main diagonal of A
- e. A row replacement operation on A does not change the determinant. A row interchange changes the sign of the determinant. A row scaling also scales the determinant.



Definitions

The eigenvalues of an $n \times n$ matrix A are the values of t that satisfy $\det(A - tI_n) = 0$.

The equation $\det(A - tI_n) = 0$ is called the **characteristic equation** of A .

The **characteristic polynomial** of A is $\det(A - tI_n)$



Example: For the matrix $A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$ find the characteristic equation of A , and use this to find the eigenvalues of A .

$$\begin{aligned} \det(A - tI_2) &= \det \begin{bmatrix} 4-t & -3 \\ 2 & -1-t \end{bmatrix} = (4-t)(-1-t) - (-3)(2) \\ &= -4 - 3t + t^2 + 6 \\ &= t^2 - 3t + 2 = \underline{(t-2)(t-1)} \end{aligned}$$

So, the eigenvalues of A are the solutions
to $\underline{(t-2)(t-1) = 0}$

$$\text{or } t = 2$$

$$t = 1$$



Example: Find the characteristic polynomial of $A = \begin{bmatrix} 4 & 2 & 0 \\ 0 & 7 & -2 \\ 8 & 8 & -3 \end{bmatrix}$

$$\begin{aligned} \det \begin{bmatrix} 4-t & 2 & 0 \\ 0 & 7-t & -2 \\ 8 & 8 & -3-t \end{bmatrix} &= (4-t)(-1)^{1+1} \det \begin{bmatrix} 7-t & -2 \\ 8 & -3-t \end{bmatrix} \\ &\quad + (2)(-1)^{1+2} \det \begin{bmatrix} 0 & -2 \\ 8 & -3-t \end{bmatrix} \\ &= (4-t) \left[(7-t)(-3-t) + 16 \right] - 2 \left[16 \right] \\ &= (4-t) \left[-21 - 4t + t^2 + 16 \right] - 32 = (4-t)(t^2 - 4t - 5) - 32 \\ &= 4t^2 - 16t - 20 - t^3 + 4t^2 + 5t - 32 \\ &= -t^3 + 8t^2 - 11t - 52 \end{aligned}$$





Definitions

The **multiplicity** of an eigenvalue λ is the number of times $(t - \lambda)$ appears as a factor in the characteristic polynomial.



Examples

The characteristic polynomial of $A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$ is $(t - 2)(t - 1)$. Each of the eigenvalues $\lambda_1 = 2$, $\lambda_2 = 1$ has multiplicity 1.

The characteristic polynomial of $A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix}$ is $(t + 1)^2(t - 3)$. The eigenvalue $\lambda_1 = -1$ has multiplicity 2, while the eigenvalue $\lambda_2 = 3$ has multiplicity 1.

A matrix with characteristic polynomial $(t - 4)^3(t + 2)^2(t - 3)$ has

- Eigenvalue $\lambda_1 = 4$ with multiplicity 3
- Eigenvalue $\lambda_2 = -2$ with multiplicity 2
- Eigenvalue $\lambda_3 = 3$ with multiplicity 1



Example: Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$. Find the eigenvalues of A , their multiplicities and a basis for each eigenspace.

$$\det \begin{bmatrix} 2-t & 1 & 0 \\ 0 & 1-t & 2 \\ 0 & 0 & 1-t \end{bmatrix} = (2-t)(1-t)(1-t) = (2-t)(1-t)^2$$

$$\lambda_1 = 2 \quad \text{multiplicity } 1$$

$$\lambda_2 = 1 \quad \text{multiplicity } 2$$



$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \quad \underline{\underline{\lambda_1 = 1}}$$

find null space $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\frac{1}{2}r_2 \rightarrow r_2} \begin{matrix} x_1 & x_2 & x_3 \\ \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$\left. \begin{array}{l} x_1 + x_2 = 0 \\ x_3 = 0 \end{array} \right\}$$

$$\left. \begin{array}{l} x_1 = -x_2 \\ x_2 = x_2 \\ x_3 = 0 \end{array} \right\}$$

Vector form: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

Basis = $\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$



$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\lambda_2 = 2$$

Null space

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

$$r_1 + r_2 \rightarrow r_2$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

$$r_2 + 2r_3 \rightarrow r_2$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\begin{matrix} x_1 & x_2 & x_3 \\ \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$x_2 = 0$$

$$x_3 = 0$$



$$x_1 = x_1$$

$$x_2 = 0$$

$$x_3 = 0$$

Vector form: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$$\text{Basis} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$



Definition: An $n \times n$ matrix A is **similar** to an $n \times n$ matrix B if $A = PBP^{-1}$ for some invertible matrix P

Note: If $A = PBP^{-1}$ then $B = QAQ^{-1}$ so B is also similar to A

Note: If $A = PBP^{-1}$ then $A^n = PB^nP^{-1}$

$$\begin{aligned} A^2 &= (PBP^{-1})(PBP^{-1}) \\ &= P B (P^{-1}P) B P^{-1} \\ &= P B B P^{-1} \\ &= P B^2 P^{-1} \end{aligned}$$

$$\begin{aligned} A &= PBP^{-1} \\ AP &= PB(P^{-1}P) \\ \boxed{AP} &= \boxed{PB} \\ P^{-1}AP &= (P^{-1}P)B \\ P^{-1}AP &= B \\ \text{Let } Q &= P^{-1} \\ B &= QAQ^{-1} \end{aligned}$$



Theorem 4

If A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues.

Caution: Similar matrices have the same characteristic polynomial and eigenvalues. If two matrices have the same characteristic polynomial and eigenvalues, that does not mean they are similar.



Theorem 4 - Proof

If $B = P^{-1}AP$, then expand the expression $B - \lambda I$:

$$\underline{B - \lambda I} = \underline{P^{-1}AP} - \lambda \underline{P^{-1}P} = \underline{P^{-1}(AP - \lambda P)} = \underline{P^{-1}(A - \lambda I)P}$$

Using the properties of determinants

$$\underline{\det(B - \lambda I)} = \underline{\det(P^{-1})} * \underline{\det(A - \lambda I)} * \underline{\det(P)}$$

But $\underline{\det(P^{-1})} = 1/\det(P)$

$$\underline{\det(B - \lambda I)} = \underline{\det(A - \lambda I)}$$



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